

MITCHEL TU

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Technical Skills

Languages & Cloud: Python, Java, C++, JavaScript/TypeScript, SQL, Swift, AWS (Lambda, Bedrock, DynamoDB, S3)

AI, ML & Frameworks: PyTorch, XGBoost, MediaPipe, ONNX, TGCN, React/Next.js, FastAPI

Systems & Infrastructure: WebSockets, WebRTC, Asyncio, ROS 2, Docker, IaC (AWS SAM), CI/CD

Education

University of California, Irvine

September 2024 – Expected June 2028

Bachelor of Science in Software Engineering – GPA: 3.70

Irvine, CA

Relevant Coursework

- Data Structures
- Algorithms Analysis
- Project Management
- Operating Systems
- Databases & Web Apps
- Database Management
- Requirements Analysis
- Computer Architecture

Experience

Software Developer Internship

August 2026 – Present

EcoServants - Ecological Servants Project

Remote

- Engineered a persistent campaign management system for a custom WordPress plugin using PHP and MySQL CRUD operations, enabling users to create, edit, duplicate, schedule, and resend bulk email campaigns.
- Designed database schemas and integrated status-tracking logic to manage draft states, automated scheduling queues, and real-time delivery analytics within the WordPress admin interface.

Undergraduate AI Researcher

July 2026 – Present

UCI Department of Statistics, Professor Weining Shen

Irvine, CA

- Built a full NBA shot-value pipeline in Python using pandas, NumPy, scikit-learn, and XGBoost to fuse play-by-play and tracking data, estimate expected points, and model counterfactual dependence of three-point attempts.
- Applied feature engineering, statistical modeling, and data analysis in Python to develop player-, team-, and game-level dependence metrics, enabling evidence-based evaluation of shot selection and offensive value in sports analytics.
- Eliminated model leakage by replacing in-sample 75/25 scoring with grouped 5-fold cross-validation in scikit-learn, reducing inflated AUC from 90% to below 60% on honest out-of-fold evaluation, through agile workflows.

Peer Learning Assistant

January 2026 – June 2026

UCI Intermediate Python Programming Course

Irvine, CA

- **Selected** as a Peer Learning Assistant to facilitate technical labs for a **Python programming** course.
- Execute technical **code reviews** for 30+ students, identifying logic errors in API integrations, Unit Testing, and third-party library implementations.

Projects

Autonomous LiDAR RC Car | [Code](#)

June 2026 - Present

- Engineered a 3-tier autonomous mobile robot integrating a custom native Swift iOS app for ARKit LiDAR streaming, ROS 2 Humble edge compute, and ESP32 firmware for real-time 3D SLAM and frontier exploration.
- Developed probabilistic log-odds voxel grid mapping using 3D DDA raycasting, eliminating redundant full-cloud recalculations by processing only incoming frame points and capping compute at 2,000 rays/frame alongside A* navigation and Yamauchi frontier detection.
- Architected a lightweight binary TCP streaming protocol in Swift/C++ for 6-DoF pose and point cloud data at 15 bytes/point (75% payload reduction vs. JSON) with ICP scan-to-map drift correction for encoder-free localization.
- Designed a distributed 10-node ROS 2 system with a 10 Hz real-time control loop, automated motor drive calibration, and a 500 ms hardware watchdog safety failsafe.

NIMBUS – Live ASL Translation Over Video Call | [Live Demo](#) • [Post](#)

March 2026

- Architected a real-time word-level ASL recognition system using MediaPipe, TGCN, and ONNX on a hybrid edge/cloud framework, achieving 15ms local gesture detection and sub-1.5s end-to-end latency.
- Engineered an event-driven serverless pipeline on AWS using Lambda, WebSocket API Gateway, and DynamoDB alongside Amazon Bedrock and Polly to translate dynamic gestures into context-aware, emotion-infused speech.

Mathnasium Chrome Extension – Instruction Workflow Automation | [Code](#)

September 2025

- Built a dashboard extension with **OCR (Tesseract.js)** and serverless functions automating instructional workflows.
- Integrated an **LLM gateway** for real-time math hint generation, reducing instructor session prep time by 30%.